PG No. 45 $C_{6h}(c)$ $6/m$ [hexagonal] (internal polar dipole)

* Harmonics for rank 0

$\vec{\mathbb{Q}}_0^{(1,1)}[q](A_g)$

** symmetry

1

** expression

$$\frac{\sqrt{3}Q_x x}{3}+\frac{\sqrt{3}Q_y y}{3}+\frac{\sqrt{3}Q_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(1,0)}[q](A_g)$

** symmetry

z

** expression

$$\frac{\sqrt{2}Q_x y}{2}-\frac{\sqrt{2}Q_y x}{2}$$

$\vec{\mathbb{G}}_1^{(1,0)}[q](E_{1g}^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{iQ_x z}{2}+\frac{Q_y z}{2}+\frac{iQ_z\left(x+iy\right)}{2}$$

$\vec{\mathbb{G}}_1^{(1,0)}[q](E_{1g}^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{iQ_x z}{2}+\frac{Q_y z}{2}-\frac{iQ_z\left(x-iy\right)}{2}$$

$\vec{\mathbb{Q}}_1^{(1,-1)}[q](A_u)$

** symmetry

z

** expression

Q_z

$\vec{\mathbb{Q}}_1^{(1,1)}[q](A_u)$

** symmetry

z

** expression

$$\frac{3\sqrt{10}Q_x xz}{10}+\frac{3\sqrt{10}Q_y yz}{10}-\frac{\sqrt{10}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_1^{(1,-1)}[q](E_{1u}^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{2}Q_x}{2}+\frac{\sqrt{2}iQ_y}{2}$$

$\vec{\mathbb{Q}}_1^{(1,1)}[q](E_{1u}^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{5}Q_x\left(2x^2+3ixy-y^2-z^2\right)}{10}-\frac{\sqrt{5}iQ_y\left(x^2+3ixy-2y^2+z^2\right)}{10}+\frac{3\sqrt{5}Q_zz\left(x+iy\right)}{10}$$

$\vec{\mathbb{Q}}_1^{(1,-1)}[q](E_{1u}^{(b)})$
** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{2}Q_x}{2}-\frac{\sqrt{2}iQ_y}{2}$$

$\vec{\mathbb{Q}}_1^{(1,1)}[q](E_{1u}^{(b)})$
** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{5}Q_x\left(2x^2-3ixy-y^2-z^2\right)}{10}+\frac{\sqrt{5}iQ_y\left(x^2-3ixy-2y^2+z^2\right)}{10}+\frac{3\sqrt{5}Q_zz\left(x-iy\right)}{10}$$

* Harmonics for rank 2
$\vec{\mathbb{Q}}_2^{(1,-1)}[q](A_g)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{6}Q_xx}{6}-\frac{\sqrt{6}Q_yy}{6}+\frac{\sqrt{6}Q_zz}{3}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](A_g)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}Q_xx\left(x^2+y^2-4z^2\right)}{14}-\frac{\sqrt{21}Q_yy\left(x^2+y^2-4z^2\right)}{14}-\frac{\sqrt{21}Q_zz\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](E_{1g}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{Q_xz}{2}+\frac{iQ_yz}{2}+\frac{Q_z\left(x+iy\right)}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](E_{1g}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{14}Q_xz\left(4x^2+5ixy-y^2-z^2\right)}{14}-\frac{\sqrt{14}iQ_yz\left(x^2+5ixy-4y^2+z^2\right)}{14}-\frac{\sqrt{14}Q_z\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](E_{1g}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{Q_xz}{2}-\frac{iQ_yz}{2}+\frac{Q_z\left(x-iy\right)}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](E_{1g}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{14}Q_x z\left(4x^2 - 5ixy - y^2 - z^2\right)}{14} + \frac{\sqrt{14}iQ_y z\left(x^2 - 5ixy - 4y^2 + z^2\right)}{14} - \frac{\sqrt{14}Q_z\left(x - iy\right)\left(x^2 + y^2 - 4z^2\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](E_{2g}^{(a)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{Q_x\left(x - iy\right)}{2} - \frac{iQ_y\left(x - iy\right)}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](E_{2g}^{(a)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{14}Q_x\left(x - iy\right)\left(3x^2 - 5ixy - 2y^2 - 2z^2\right)}{28} + \frac{\sqrt{14}iQ_y\left(x - iy\right)\left(2x^2 - 5ixy - 3y^2 + 2z^2\right)}{28} + \frac{5\sqrt{14}Q_z z\left(x - iy\right)^2}{28}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](E_{2g}^{(b)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{Q_x\left(x + iy\right)}{2} + \frac{iQ_y\left(x + iy\right)}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](E_{2g}^{(b)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{14}Q_x\left(x + iy\right)\left(3x^2 + 5ixy - 2y^2 - 2z^2\right)}{28} - \frac{\sqrt{14}iQ_y\left(x + iy\right)\left(2x^2 + 5ixy - 3y^2 + 2z^2\right)}{28} + \frac{5\sqrt{14}Q_z z\left(x + iy\right)^2}{28}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](A_u)$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{6}Q_x yz}{2} - \frac{\sqrt{6}Q_y xz}{2}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](E_{1u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{Q_x\left(xy + iy^2 - iz^2\right)}{2} - \frac{Q_y\left(x^2 + ixy - z^2\right)}{2} + \frac{iQ_z z\left(x + iy\right)}{2}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](E_{1u}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{Q_x\left(xy-iy^2+iz^2\right)}{2}-\frac{Q_y\left(x^2-ixy-z^2\right)}{2}-\frac{iQ_zz\left(x-iy\right)}{2}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](E_{2u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{iQ_xz\left(x-iy\right)}{2}+\frac{Q_yz\left(x-iy\right)}{2}-\frac{iQ_z\left(x-iy\right)^2}{2}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](E_{2u}^{(b)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$-\frac{iQ_xz\left(x+iy\right)}{2}+\frac{Q_yz\left(x+iy\right)}{2}+\frac{iQ_z\left(x+iy\right)^2}{2}$$

* Harmonics for rank 3
$\vec{\mathbb{G}}_3^{(1,0)}[q](A_g)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{3}Q_xy\left(x^2+y^2-4z^2\right)}{4}+\frac{\sqrt{3}Q_yx\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](B_g,1)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{\sqrt{30}Q_xz\left(x-y\right)\left(x+y\right)}{8}+\frac{\sqrt{30}Q_yxyz}{4}+\frac{\sqrt{30}Q_zx\left(x^2-3y^2\right)}{8}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](B_g,2)$
** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{30}Q_xxyz}{4}+\frac{\sqrt{30}Q_yz\left(x-y\right)\left(x+y\right)}{8}-\frac{\sqrt{30}Q_zy\left(3x^2-y^2\right)}{8}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](E_{1g}^{(a)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$$

** expression

$$\frac{iQ_xz\left(x^2-10ixy+11y^2-4z^2\right)}{8}-\frac{Q_yz\left(11x^2+10ixy+y^2-4z^2\right)}{8}-\frac{iQ_z\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{8}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](E_{1g}^{(b)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$-\frac{iQ_xz\left(x^2+10ixy+11y^2-4z^2\right)}{8}-\frac{Q_yz\left(11x^2-10ixy+y^2-4z^2\right)}{8}+\frac{iQ_z\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{8}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](E_{2g}^{(a)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{10}Q_x\left(x-iy\right)\left(xy-iy^2+2iz^2\right)}{8} - \frac{\sqrt{10}Q_y\left(x-iy\right)\left(x^2-ixy-2z^2\right)}{8} - \frac{\sqrt{10}iQ_zz\left(x-iy\right)^2}{4}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](E_{2g}^{(b)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{10}Q_x\left(x+iy\right)\left(xy+iy^2-2iz^2\right)}{8} - \frac{\sqrt{10}Q_y\left(x+iy\right)\left(x^2+ixy-2z^2\right)}{8} + \frac{\sqrt{10}iQ_zz\left(x+iy\right)^2}{4}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](A_u)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{\sqrt{15}Q_xxz}{5} - \frac{\sqrt{15}Q_yyz}{5} - \frac{\sqrt{15}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](A_u)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{5Q_xxz\left(3x^2+3y^2-4z^2\right)}{12} - \frac{5Q_yyz\left(3x^2+3y^2-4z^2\right)}{12} + \frac{Q_z\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](B_u,1)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{6}Q_xxy}{2} + \frac{\sqrt{6}Q_y\left(x-y\right)\left(x+y\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](B_u,2)$
** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{6}Q_x\left(x-y\right)\left(x+y\right)}{4} - \frac{\sqrt{6}Q_yxy}{2}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](B_u,1)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{10}Q_xxy\left(15x^2-13y^2-6z^2\right)}{24} - \frac{\sqrt{10}Q_y\left(3x^4-21x^2y^2+3x^2z^2+4y^4-3y^2z^2\right)}{24} + \frac{7\sqrt{10}Q_zyz\left(3x^2-y^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](B_u,2)$
** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{10}Q_x\left(4x^4-21x^2y^2-3x^2z^2+3y^4+3y^2z^2\right)}{24}+\frac{\sqrt{10}Q_yxy\left(13x^2-15y^2+6z^2\right)}{24}+\frac{7\sqrt{10}Q_zxz\left(x^2-3y^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](E_{1u}^{(a)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{5}Q_x\left(3x^2+2ixy+y^2-4z^2\right)}{20}-\frac{\sqrt{5}iQ_y\left(x^2-2ixy+3y^2-4z^2\right)}{20}+\frac{2\sqrt{5}Q_zz\left(x+iy\right)}{5}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](E_{1u}^{(a)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{3}Q_x\left(4x^4+5ix^3y+3x^2y^2-27x^2z^2+5ixy^3-30ixyz^2-y^4+3y^2z^2+4z^4\right)}{24}$$
$$+\frac{\sqrt{3}iQ_y\left(x^4+5ix^3y-3x^2y^2-3x^2z^2+5ixy^3-30ixyz^2-4y^4+27y^2z^2-4z^4\right)}{24}-\frac{5\sqrt{3}Q_zz\left(x+iy\right)\left(3x^2+3y^2-4z^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](E_{1u}^{(b)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{5}Q_x\left(3x^2-2ixy+y^2-4z^2\right)}{20}+\frac{\sqrt{5}iQ_y\left(x^2+2ixy+3y^2-4z^2\right)}{20}+\frac{2\sqrt{5}Q_zz\left(x-iy\right)}{5}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](E_{1u}^{(b)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{3}Q_x\left(4x^4-5ix^3y+3x^2y^2-27x^2z^2-5ixy^3+30ixyz^2-y^4+3y^2z^2+4z^4\right)}{24}$$
$$-\frac{\sqrt{3}iQ_y\left(x^4-5ix^3y-3x^2y^2-3x^2z^2-5ixy^3+30ixyz^2-4y^4+27y^2z^2-4z^4\right)}{24}-\frac{5\sqrt{3}Q_zz\left(x-iy\right)\left(3x^2+3y^2-4z^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](E_{2u}^{(a)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{2}Q_xz\left(x-iy\right)}{2}-\frac{\sqrt{2}iQ_yz\left(x-iy\right)}{2}+\frac{\sqrt{2}Q_z\left(x-iy\right)^2}{4}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](E_{2u}^{(a)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{30}Q_xz\left(x-iy\right)\left(5x^2-7ixy-2y^2-2z^2\right)}{24}+\frac{\sqrt{30}iQ_yz\left(x-iy\right)\left(2x^2-7ixy-5y^2+2z^2\right)}{24}-\frac{\sqrt{30}Q_z\left(x-iy\right)^2\left(x^2+y^2-6z^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](E_{2u}^{(b)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{2}Q_x z\left(x+iy\right)}{2}+\frac{\sqrt{2}iQ_y z\left(x+iy\right)}{2}+\frac{\sqrt{2}Q_z\left(x+iy\right)^2}{4}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](E_{2u}^{(b)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{30}Q_x z\left(x+iy\right)\left(5x^2+7ixy-2y^2-2z^2\right)}{24}-\frac{\sqrt{30}iQ_y z\left(x+iy\right)\left(2x^2+7ixy-5y^2+2z^2\right)}{24}-\frac{\sqrt{30}Q_z\left(x+iy\right)^2\left(x^2+y^2-6z^2\right)}{24}$$